europass Curriculum vitae

PERSONAL INFORMATION

PhD student in Biomolecular Sciences focused on the analysis of complex biomedical data, with particular emphasis on omics. Proposer of the JoinTheLab project, which aims to combine technical expertise in the analysis of complex NGS data with an entrepreneurial vision targeted at the digitalisation of laboratory processes. Committed to defining new reproducibility standards for the biotech sector through the technology transfer of academic research.

EDUCATION

November 2024 – Ongoing

PhD in Biomolecular Sciences – Laboratory of Computational Modeling

University of Trento, Department of CIBIO, Trento (Italy)

- Implemented an innovative rank-based regression method for explainable prediction of continuous clinical variables from omics datasets of various nature, including transcriptomics, metagenomics and single cell transcriptomics.
- Pre-processing and analysis of omics datasets, obtaining biologically relevant insights through the application of explainable machine learning methods.
- Presented my work at various conferences, including the CIBB 2025 and the DS4HB conferences.

September 2021 – October 2023

Master's Degree in Quantitative and computational biology

University of Trento, Trento (Italy)

Master thesis: *"Synchronization of neuronal activity in the antennal lobe of the honeybee during sleep"*, Supervisor Prof. A. Haase. Final grade 110/110 cum laude.

Relevant Coursework:

- Bioinformatics.
- Machine Learning.
- Mathematical Modeling.
- Computational Biophysics.
- Function and plasticity of the Central Nervous System.

September 2018 – September 2021

Bachelor's Degree in Computer Science

University of Trento, Trento (Italy)

Bachelor thesis: *"Analysis of trascriptomic RNA-seq data from polisomial and total fraction from an epithelial cancer cellular line"*. Supervisor Prof. A. Inga. Final grade 108/110

Relevant Coursework:

- Molecular biology.
- Genetics.

EXPERIENCE

November 2023 – November 2024

Research Fellow – PREDICTive Cardio-metabolic transcriptOMIC trajectorieS in the Barilla Offspring Follow-up STUDY: The PREDICT-OMICS Study

University of Trento, Department of CIBIO, Trento (Italy)

- Implemented a fully automatic computational pipeline to process microarray data from the Barilla Offspring study.
- The pipeline, written in python and R, processes raw data and applies sequentially statistical and machine learning (both classification and regression) state-of-the-art methods to infer genes predictive for disease relevant clinical measures.
- Analysed pipeline results to gain biological insights from it.
- Results have been presented as a poster at the BITS 2024 conference.

March 2023 – August 2023

Intern – Computational modelling and simulation of the antennal lobe of the honeybee

University of Trento, Department of CiMeC, Rovereto (Italy)

- Implemented a gpu-accelerated computational model of the antennal lobe of the honeybee in python using the genn library.
- Implemented a machine learning pipeline to analyse both experimental and model data.
- The code is accessible at https://github.com/giacThePhantom/genn-network-model.
- The work resulted in a publication in *Neural Networks* (2025), see Publications.

March 2021 – August 2021

Intern – Analysis of transcriptomic RNA-seq data

University of Trento, Department of CIBIO, Trento (Italy)

- Implemented a pipeline to process RNA-seq cancer data on a remote server.
- Analysed processed data using a combination of R and python to gain biological insights.
- The work is accessible at https://github.com/giacThePhantom/Tesi.

MENTORING AND TEACHING

September 2025 – December 2025

Teaching Assistant – Computer Science course

University of Trento, Department of CIBIO, Trento (Italy)

Supported the teaching of the Computer Science course, assisting students in learning programming and solving algorithmic problems.

Thesis co-supervision

Provided technical and scientific mentoring to two students during their experimental thesis work. Coordinated the planning of their analyses and trained them in the use of bioinformatics tools, supervising the progress of the projects up to the final dissertation.

PUBLICATIONS

2025 *"Neuronal correlates of sleep in honey bees"*. Neural Networks, Volume 189, 2025, 107575. DOI: 10.1016/j.neunet.2025.107575.

PERSONAL PROJECTS

Personal coursework collection

Wrote in LaTeX all the material regarding the content of my university courses, making it accessible at https://github.com/giacThePhantom. This has provided easily accessible quality material for my colleagues and it has encouraged some people to join this project, creating a small community of maintainers.

PERSONAL SKILLS

Mother tongue: Italian

Other languages

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken interaction	Spoken production	
English	C1	C1	C1	C1	C1

Levels: A1 and A2: Basic user – B1 and B2: Independent user – C1 and C2: Proficient user
Common European Framework of Reference for Languages

Digital competences

SELF-ASSESSMENT				
Information Processing	Communication	Content creation	Safety	Problem solving
Proficient user	Proficient user	Proficient user	Proficient user	Proficient user

Digital competences - Self-assessment grid

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Programming skills
- Python (numpy, pandas, scikit-learn, pytorch).
- R.
- C/C++.
- Java.
- Bash.
- Matlab.
- Javascript.
- Linux systems.
- Git.
- LaTeX.